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ORIGINAL PAPER

Robust Classification of Urban Sounds in Noisy Environments: A Novel Approach Using SPWVD-MFCC and Dual-Stream Classifier

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Abstract
Urban sound classification is essential for effective sound monitoring and mitigation strategies, which are critical to addressing the negative impacts of noise pollution on public health. While existing methods predominantly rely on Short-Term Fourier Transform (STFT)-based features like Mel-Frequency Cepstral Coefficients (MFCC), these approaches often struggle to identify the dominant sound in noisy environments. This gap in robustness limits the practical deployment of such systems in real-world urban settings, where noise levels are unpredictable and variable. Here, we introduce Smoothed Pseudo-Wigner–Ville Distribution-based MFCC (SPWVD-MFCC), a novel feature that merges SPWVD's high time–frequency resolution with MFCC's human-like auditory sensitivity. We further propose a dual-stream ResNet50-CNN-LSTM architecture to classify these features. Comprehensive experiments conducted on UrbanSound8K, UrbanSoundPlus, and DCASE2016 datasets demonstrate that the proposed SPWVD-MFCC significantly improves classification accuracy in noisy conditions, with an enhancement of up to 37.2% over traditional STFT-based methods and better robustness than existing approaches. These results indicate that the proposed approach addresses a critical gap in urban sound classification by providing enhanced robustness in low-SNR environments. This advancement improves the reliability of urban noise monitoring systems and contributes to the broader goal of creating healthier urban living environments by enabling more effective noise-control strategies.

Keywords Urban sound classification · Sound classification in noisy environments · Wigner–Ville Distribution · SPWVD-MFCC · Dual-stream classifier

1 Introduction

Urban sound classification is a vital component of urban noise monitoring, offering categorised sound data to relevant authorities to support the development of targeted noise-control strategies. These strategies can effectively reduce noise levels in specific areas, minimising the negative impacts of noise pollution on human health and well-being. Previous studies [1–8] have highlighted that excessive noise levels can lead to various health issues, including hearing loss, sleep disturbances, and cardiovascular diseases. Consequently, classifying urban sounds is essential for ensuring a healthier living environment.

Existing research in urban sound classification primarily relies on short-term Fourier transform (STFT)-type spectrogram and its derived features, such as Mel-Frequency Cepstral Coefficients (MFCCs), achieving notable success. MFCC maps the linear frequency bins of STFT to a nonlinear Mel scale to simulate the perception of the human ear [9]. However, a common limitation in these studies is that the test sets are often randomly split from the datasets, making the testing set closely resemble the training data. In contrast, real-world urban environments consist of complex mixtures of sounds from multiple sources, which challenge the noise robustness of these systems and raise concerns about their practical deployment in nonideal conditions.

Noise is often unpredictable and can occur at any frequency, adversely affecting classification accuracy. Higher time–frequency resolution in features is crucial for improving classification performance. However, STFT struggles to effectively balance time and frequency resolution, often necessitating trade-offs that limit its robustness in noisy

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environments [10]. Similarly, other linear time–frequency analysis methods depending on windowing, such as continuous wavelet transform, Gabor transform, and S transform, cannot improve time and frequency resolutions simultaneously. The nonlinear methods, such as adaptive STFT, have insufficient robustness and are hard to adapt to the low-SNR scenarios [11]. Research across various signal processing fields has demonstrated that the bilinear (also called quadratic) method Wigner–Ville Distribution (WVD) offers superior time and frequency resolution simultaneously [11, 12], making it more noise-resistant. Smoothed Pseudo-WVD (SPWVD), a derivative of WVD, significantly addresses the cross-term interference commonly found in WVD, enhancing the accuracy of time–frequency analysis [12]. Despite these advantages, SPWVD has not been fully leveraged in urban sound classification tasks, leaving their potential for noise resistance largely unexplored.

This study aims to enhance the robustness of urban sound classification in noisy environments. Specifically, it improves the ability to distinguish the dominant type of urban sound from a highly noise-polluted acoustic environment with low SNR. It is an area rarely addressed. To achieve this, we introduce SPWVD-based MFCC (SPWVD-MFCC). The development of SPWVD-MFCC is guided by two key insights: first, SPWVD offers superior time–frequency resolution compared to STFT and fewer cross-terms than WVD, enabling its noise robustness; second, traditional MFCC has demonstrated excellent performance in many studies by effectively simulating human auditory perception [13–16]. As a result, SPWVD-MFCC is anticipated to deliver greater robustness in low-SNR environments than traditional MFCC. Additionally, we propose employing a dual-stream ResNet50-CNN-LSTM architecture for feature classification. Both ResNet50 and CNN-LSTM have been widely recognised for offering outstanding classification accuracy. In the dual-stream classifier, ResNet50 processes the frequency and time information of the features together; the CNN-LSTM first applies CNN to process frequency information only, followed by LSTM which exclusively handles temporal information. The final decision is based on both streams. We then conduct a comparative analysis of the accuracy of STFT, SPWVD, STFT-MFCC, and SPWVD-MFCC across various classifiers, including ResNet50, CNN-LSTM, MobileViT-s, and ResNet50-CNN-LSTM, presenting detailed experimental results to support our proposal. The main contributions of this paper are as follows:

- A novel feature, namely SPWVD-MFCC, is proposed combining the high time–frequency resolution of SPWVD with the auditory modelling of MFCC. This approach significantly improves classification accuracy in noisy urban environments, addressing a major limitation of traditional STFT-based methods.
- A dual-stream ResNet50-CNN-LSTM classifier is proposed, leveraging spatial and temporal feature extraction capabilities. By integrating the strengths of ResNet50 and CNN-LSTM, this architecture delivers superior classification performance, particularly in complex, low-SNR scenarios.
- Extensive experiments conducted on UrbanSound8K, the newly created UrbanSoundPlus, and DCASE2016 datasets demonstrate the effectiveness of the proposed SPWVD-MFCC and dual-stream architecture. The results show consistent improvements in classification accuracy, highlighting the robustness and adaptability of the system in diverse and noisy urban soundscapes.

The structure of this paper is organised as follows. Section 2 provides a detailed overview of related works. Section 3 comprehensively introduces the dataset preparation, data pre-processing, features, classifiers, and experiment settings. Section 4 discusses the experiments, result analyses, performance comparisons with other studies, and limitations. Finally, Sect. 5 concludes this paper.

2 Related Work

Sound classification is a well-explored field, with numerous techniques developed to enhance accuracy by extracting features from raw audio slices and feeding them into classifiers. Among the most widely used features is the STFT-type spectrogram, which serves as a foundation for many derived features such as Mel-STFT, log-Mel-STFT, and MFCC and often acts as a benchmark for comparison.

For instance, Ananya et al. compared the performance of STFT, MFCC, and audio embeddings for indoor–outdoor binary classification on the DCASE2020 dataset. They found that STFT achieved higher accuracies across various CNN models, with ShuffleNet reaching up to 93.1% accuracy, outperforming MFCC and audio embeddings. [17] Meghanani et al. explored sound classification in the context of Alzheimer's dementia recognition, discovering that MFCC features classified by CNN-LSTM yielded the highest accuracy compared to log-Mel-STFT and ComParE features [18]. Similarly, Peng et al. demonstrated that MFCC features classified by CNN-LSTM performed better than ResNet-18, ShuffleNetV1, and Gaussian Mixture Models, with MFCC outperforming other features such as Perceptual Linear Predictive Cepstral Coefficients (PLPCC) and Gammatone-scaled coefficients [16]. Beyond using single features and classifiers, multiple studies have explored multi-feature and multi-stream approaches to enhance classification accuracy.

For example, Su et al. proposed a dual-stream CNN classifier where one stream processed log-Mel-STFT alongside Chroma, Spectral Contrast, and Tonnetz features, while the other stream processed MFCC with additional features, achieving a higher accuracy of 97.2% on UrbanSound8K compared to single-stream approaches [19]. Dong et al. introduced another dual-stream CNN model that separately processed log-Mel-STFT and raw audio through different network structures, achieving 94.97% accuracy on UrbanSound8K [20].

Despite extensive testing of various features, classifiers, and system structures to enhance sound classification accuracy, most existing systems heavily rely on the STFT. STFT involves segmenting a signal into multiple windows, applying a window function to each, and then analysing them using the fast Fourier transform (FFT) algorithm. To reduce information loss due to windowing, overlapping segments are commonly used. However, researchers like Scholl [10], a fundamental limitation of STFT is the time–frequency trade-off: short windows provide high time resolution but poor frequency resolution due to fewer samples per window, while long windows improve frequency resolution at the cost of time resolution. This trade-off prevents the linear STFT from fully capturing the time–frequency details of signals, resulting in reduced performance when processing nonstationary signals with rapidly changing frequency components. Similarly, other linear time–frequency analysis methods that rely on windowing, such as continuous wavelet transform, Gabor transform, and S transform, face the same limitation of balancing time and frequency resolutions. Nonlinear methods, like adaptive STFT, have attempted to enhance resolution; however, as highlighted in the study [11], they often lack robustness and struggle to adapt to low-SNR conditions.

Unlike linear or nonlinear methods, the bilinear Wigner–Ville Distribution (WVD) calculates time–frequency distributions by applying a Fourier transform to the signal's autocorrelation function. This unique approach enables WVD to achieve high resolution in both time and frequency domains simultaneously, without the constraints of a fixed window [11, 12]. Studies by Ou et al. [21] and Su et al. [22] have demonstrated the superior performance of WVD in various applications. In sound classification, WVD has been explored through multiple studies. For instance, Hinrichs et al. proposed extracting MFCC features from different configurations, including STFT, Stockwell, WVD, and STFT-filtered WVD, for speech command recognition using CNN. Their results showed that, although STFT alone outperformed WVD, STFT-filtered WVD, and Stockwell features individually, combining all four feature types led to better classification accuracy than using STFT alone. [23] However, this evaluation was conducted on clean audio, and thus, the robustness of these methods in noisy conditions with unstable and rapidly varying frequency components remains untested. To address this gap, we replicated their approach to assess the classification robustness in noisy environments compared to our proposed method, as discussed in Sect. 4.5.

While WVD offers high-resolution time–frequency analysis, it suffers from cross-term interference, which can distort the time–frequency representation and lead to misleading interpretations. Smoothed Pseudo-WVD (SPWVD) addresses this issue by effectively reducing cross-term interference, resulting in cleaner and more accurate distributions. Various researchers have demonstrated this; for instance, Xu et al. demonstrated SPWVD's superior performance in analysing geographical signals, showing that it provides better-focusing effects and higher resolution than STFT [11]. Similarly, SPWVD has been proven to outperform Pseudo-WVD (PWVD), WVD, and STFT in applications such as seismic signal processing and emotion classification in electroencephalogram signals, offering a more precise representation of time–frequency coefficients [24, 25].

To address the challenge of urban sound classification in noisy environments, we propose the SPWVD-MFCC method, which combines the high time–frequency resolution of SPWVD with the auditory sensitivity of MFCC. This integration is designed to enhance robustness in low-SNR conditions, outperforming traditional MFCC. Experimental results confirm that pairing SPWVD-MFCC with a dual-stream ResNet50-CNN-LSTM classifier significantly boosts classification accuracy in noisy settings, thereby validating the effectiveness of our approach.

3 Methodology

This section introduces the core information of the experiments in the preparation stage, including the dataset preparation and data pre-processing, the feature extraction approach, the classifier architecture, and the experiment settings.

3.1 Dataset Preparation and Data Pre-Processing

This study uses the UrbanSound8K dataset [26], a widely recognised benchmark for urban sound classification. UrbanSound8K consists of 8,732 labelled audio slices spanning ten categories commonly found in urban environments, totalling 8 h and 45 min of audio. The categories include Air conditioners, Car horns, Children playing, Dog bark, Drilling, Engine idling, Gunshots, Jackhammer, Sirens, and Street music. Originally sampled at 44.1 kHz, the dataset is down-sampled to 8 kHz to match the real-world operational conditions of the target classification system, which relies on a wireless acoustic sensor network sampling sounds at 8 kHz [27]. For preprocessing, audio segments are concatenated by class and reviewed to remove silent or mislabelled segments, resulting in a refined dataset with 7 h, 12 min, and 41 s of audio.

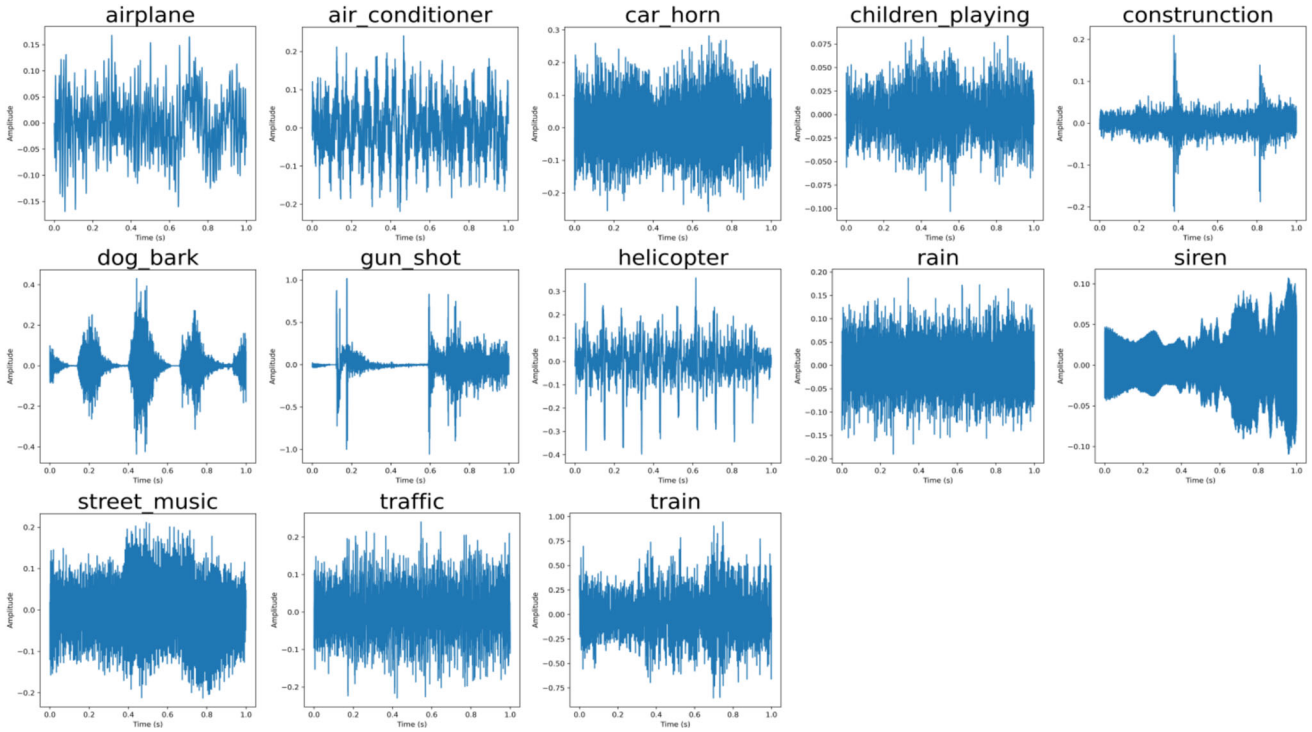


Fig. 1 Example plots of the classes in UrbanSoundPlus

To better align with real-world scenarios and optimise category coverage, a new dataset named UrbanSoundPlus is created. This dataset incorporates selected categories from UrbanSound8K and royalty-free sound effects from online sources (e.g. https://pixabay.com/sound-effects/). In UrbanSoundPlus, the Drilling and Jackhammer categories are merged into a single Construction category, reflecting their typical co-occurrence in urban construction sites. Traffic replaces the Engine idling category to encompass a broader range of road noises. Additionally, four new categories, namely Aeroplane, Helicopter, Rain, and Train, are introduced to provide a more comprehensive representation of urban sounds. Following the same refinement process, UrbanSoundPlus includes 7 h, 6 min, and 39 s of audio across the updated 13 categories: Aeroplanes, Air conditioners, Car horns, Children playing, Construction, Dog bark, Gunshots, Helicopter, Rain, Siren, Street music, Traffic, and Train. The 1-s audio examples of UrbanSoundPlus are shown in Fig. 1. These waveforms reflect the physical characteristics of different sound categories in the time domain, but it is difficult to intuitively distinguish between different sound categories based solely on the waveform graphs. Therefore, appropriate time–frequency features are crucial for accurate classification. The right features help extract the essential information of the sound, compensating for the limitations of time-domain waveforms, thereby significantly improving the performance of the classification model.

White Gaussian Noise (WGN) is mixed with clean audio signals to evaluate the classification system's robustness. WGN, characterised by random fluctuations with equal intensity across all frequencies, simulates real-world noise conditions that could impact system performance. The noise mixing is done on a 1.024-s basis, aligning with the input length required by the classifier [16] to maintain accurate SNR levels for each sample. The SNR is calculated using the formula:

$$\mathrm{SNR} = 10\log_{10}\frac{P_s}{P_n}, \tag{1}$$

thus,

$$P_n = \frac{P_s}{10^{\frac{\mathrm{SNR}}{10}}}, \tag{2}$$

where P_s is the power of the clean audio segment and P_n is the expected power of the WGN segment. The final mixed audio segment is expressed as:

$$x = s + N \times \sqrt{\frac{P_n}{P_N}}, \tag{3}$$

where x is the mixed audio segment, s is the clean audio segment, N is the generated WGN segment, P_n is the expected power of the added WGN segment, and P_N is the real power of the generated WGN segment. Given that the system aims to

classify the dominant sound in noisy environments, the lowest SNR used in this study is set at 0 dB, as negative SNRs would cause the clean signal to be overshadowed by noise. Consequently, datasets mixed with WGN at SNR levels of 0, 4, 8, 12, 16, and 20 dB are generated for the experiments.

The dataset undergoes several pre-processing steps to prepare it for feature extraction. First, the data are rescaled so that its maximum absolute value is 1, aligning with the real-world conditions of the wireless acoustic sensor network, which outputs data normalised to a maximum value of 1. This normalisation step ensures the model effectively processes and classifies sound data under realistic conditions. Next, the audio samples are split into training, validation, and testing sets in an 8:1:1 ratio. To maintain the integrity of the training and validation processes, WGN mixing is applied only to the testing set, as it is intended explicitly for robustness evaluation. Finally, Z-score normalisation is applied: the training set's mean and standard deviation are calculated and subsequently used to normalise the validation and testing sets, ensuring consistent scaling across all datasets.

3.2 Features

This study examines the performance of STFT, STFT-MFCC, SPWVD, and SPWVD-MFCC features, using STFT, STFT-MFCC, and SPWVD as benchmarks to evaluate the effectiveness of SPWVD-MFCC. Previous research [11, 12, 24, 25] has consistently shown that SPWVD provides superior signal representation compared to STFT and WVD, thanks to its enhanced time–frequency resolution. In this study, we propose SPWVD-MFCC, which combines the strengths of SPWVD with MFCC's proven efficiency in capturing auditory features, as seen in traditional STFT-MFCC applications. This section outlines the key characteristics of these features and their corresponding formulas.

3.2.1 Smoothed Pseudo-Wigner–Ville Distribution and its Derived MFCC

The Smoothed Pseudo-Wigner–Ville Distribution (SPWVD) is an advanced time–frequency representation derived from the Wigner–Ville Distribution (WVD). WVD is known for its bilinear properties, offering excellent capabilities in capturing instantaneous frequency changes, making it highly effective for analysing nonstationary signals with rapidly varying frequency components. The WVD is defined as:

$$W_x(t,\,v) = \int_{-\infty}^{+\infty} x\left(t+\frac{\tau}{2}\right)x^*\left(t-\frac{\tau}{2}\right)\mathrm{e}^{-\mathrm{j}2\pi v\tau}\mathrm{d}\tau, \tag{4}$$

where x is the signal, $x\left(t+\frac{\tau}{2}\right)x^*\left(t-\frac{\tau}{2}\right)$ is the instantaneous autocorrelation function, t is the timestamp index, and v is the frequency bin index. Although WVD provides high resolution, it suffers from cross-term interference, which can reduce distribution accuracy by introducing false components, as illustrated by comparing Fig. 2A, B from Auger et al. [28]. To address this, SPWVD introduces smoothing and windowing techniques to effectively suppress cross-terms, enhancing robustness and making it valuable for analysing nonlinear and time-varying signal characteristics. Cohen's SPWVD is defined as:

$$\begin{aligned}&\mathrm{SPW}_x(t,\,v)\\ &= \int_{-\infty}^{+\infty} h(\tau)\int_{-\infty}^{+\infty} g(s-t)\,x\left(t+\frac{\tau}{2}\right)x^*\left(t-\frac{\tau}{2}\right)\mathrm{d}s\mathrm{e}^{-\mathrm{j}2\pi v\tau}\mathrm{d}\tau,\end{aligned} \tag{5}$$

where h and g are Hamming windows applied in frequency and time domains, respectively, to smooth the representation. Figure 2C demonstrates the effectiveness of SPWVD in reducing cross-terms compared to WVD.

Building on the success of STFT-MFCC, which has shown superior accuracy over STFT alone in numerous studies, we propose SPWVD-MFCC to leverage SPWVD's high time–frequency resolution. SPWVD-MFCC integrates SPWVD with MFCC, which uses the Mel scale to mimic human auditory perception, enhancing the representation of auditory features. The conversion from linear frequency to Mel scale frequency is given by:

$$\mathrm{Mel}(f) = 2595\log_{10}\left(1+\frac{f}{700}\right), \tag{6}$$

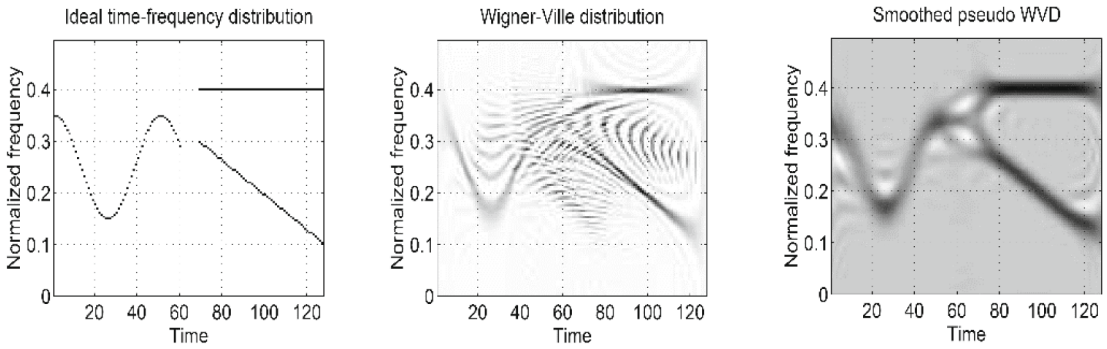
where f is the linear frequency in the time–frequency distribution. Mel filter banks filter the sound signal according to Mel scale frequencies, replicating human hearing characteristics and reducing dimensionality by mapping linear frequencies to more perceptually relevant bands. The SPWVD-MFCC is represented by:

$$W_{\mathrm{MFCC}(t,n)} = \sum_{m=0}^{M-1}\ln\left(\sum_{v=0}^{V-1} W_x(t,\,v)H_m(v)\right)\cos\left(\frac{\pi n(m+0.5)}{M}\right), \tag{7}$$

where $W_x(t,\,v)$ is the SPWVD, $H_m(v)$ is a Mel filter bank that maps the v-th frequency bin to the m-th Mel frequency, V is the total number of frequency bins, M is the total number of Mel filter banks, t is the window index, and n is the MFCC index. The block diagram in Fig. 3 illustrates the extraction process for SPWVD-MFCC.

3.2.2 Short-Term Fourier Transform and its Derived MFCC

The STFT and its derived MFCC are benchmark features to compare with SPWVD-MFCC. STFT is one of the most popular techniques in sound signal processing, and it is known for



(A) Ideal time-frequency distribution (B) Wigner-Ville Distribution (WVD) (C) Smoothed Pseudo WVD (SPWVD)

Fig. 2 Comparison of **A** ideal time–frequency distribution, **B** Wigner–Ville Distribution (WVD), and **C** Smoothed Pseudo-WVD (SPWVD)

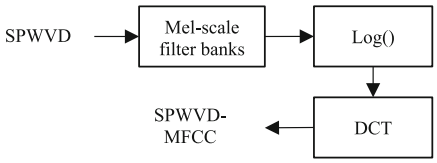


Fig. 3 Block diagram for computing SPWVD-MFCC

its effectiveness in various classification tasks. It divides the signal into overlapping windows, multiplies each window by a window function, and then applies the FFT algorithm. This generates an STFT spectrogram by concatenating the results from all windows. To minimise information loss, overlapping segments are used. The STFT is defined by:

$$S_x(t,\, v) = \left| \int_{-\infty}^{+\infty} x(u) h^*(u-t) e^{-j2\pi vu} \mathrm{du} \right|^2, \tag{8}$$

where x is the signal, h^* is the complex conjugate of the window function, t is the timestamp index, and v is the frequency bin index. STFT-MFCC follows the same extraction routines as SPWVD-MFCC, as described in Sect. 3.2.1.

3.2.3 Feature Extraction

Figure 4 visualises SPWVD, STFT, and their derived MFCCs on a decibel (dB) scale for clearer comparison. Panels (A-C) illustrate features from clean dog-bark audio, while panels (D-F) depict features from dog-bark audio mixed with WGN at an SNR of 0 dB. Each audio segment contains 8192 samples at an 8 kHz sample rate, corresponding to 1.024 s—this standardised length ensures consistency in feature extraction for the classifier. Due to the real-valued nature of the signals, only half of the frequency components from SPWVD and STFT are used, resulting in 256 frequency bins for classification from a total of 512. SPWVD features are extracted using the Python module "tftb" [29] with a step length of 32 samples, producing features with 256 timestamps. STFT features are extracted using the "torchaudio" module [30] with a hop length of 32 samples and a window length of 512 samples to maintain consistent resolution. The MFCCs are derived by mapping the linear frequencies to the Mel scale and generating 16 static cepstral coefficients. The coefficients' first and second derivatives (delta and delta-delta) are concatenated to form feature segments with 48 coefficients per timestamp.

3.3 Benchmark Classifiers

This study utilises ResNet50, CNN-LSTM, and MobileViT-s as benchmark classifiers for evaluating STFT, SPWVD, and their derived MFCC features. ResNet50 is a pure CNN model, while CNN-LSTM and MobileViT-s are CNN-based hybrid models. The literature review indicates that CNNs are the most widely used architecture in urban sound classification due to their strength in processing spatial information [31]. CNNs employ convolutional layers to analyse spatial features such as edges, textures, and patterns in input data. ResNet50, proposed by He et al. [32], is known for using residual connections that facilitate efficient gradient backpropagation, enhancing training depth and performance. This model has consistently demonstrated vital accuracy in image classification tasks across various studies [25, 33–35]. In our study, ResNet50 was selected due to its balance between model complexity and computational efficiency. Compared to deeper variants like ResNet101 or ResNet152, ResNet50 provides sufficient feature extraction capability while maintaining manageable computational cost, making it well-suited for our experimental setup.

The CNN-LSTM model integrates the spatial processing capabilities of CNNs with the temporal sequence handling of LSTMs, making it suitable for sound classification, which is rich in temporal information. LSTM, introduced by Hochreiter and Schmidhuber [36], incorporates gating mechanisms that effectively manage information flow in long sequences.

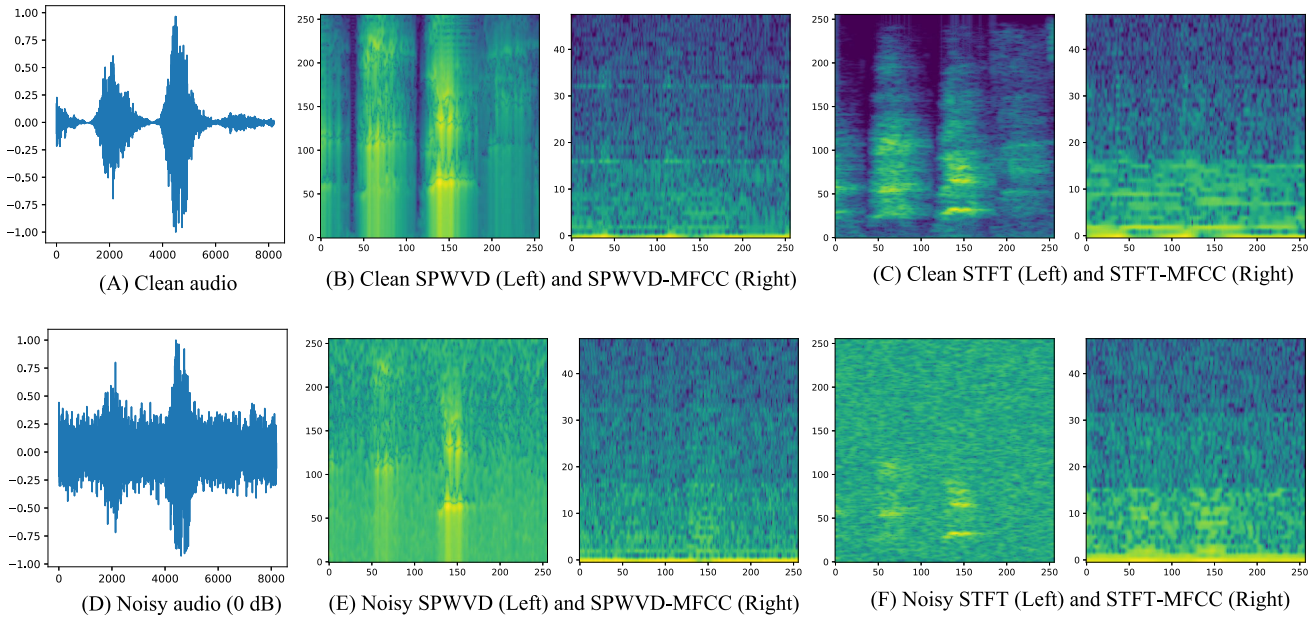
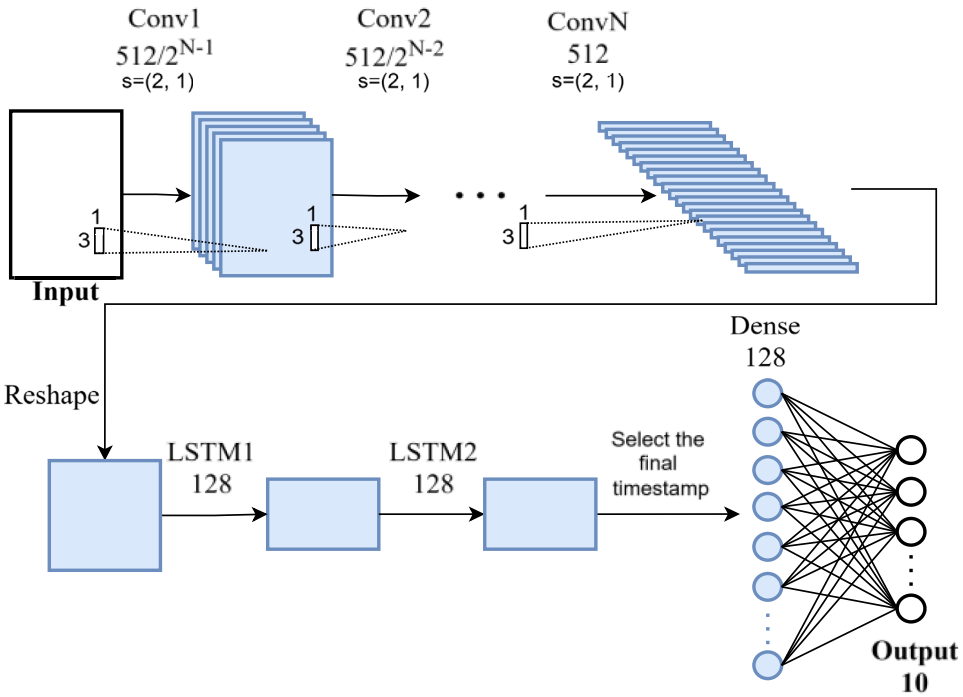


Fig. 4 The waveforms and features of clean dog-bark audio and WGN-mixed dog-bark audio with a SNR of 0 dB. In decibel-scaled figure blocks (B, C, E, and F), the left figures are SPWVD and STFT, and the right figures show the derived MFCCs

Fig. 5 Structure of the CNN-LSTM



The CNN-LSTM model employed in this study was proposed in the study [16], as depicted in Fig. 5, which processes frequency information exclusively through convolutional layers, reducing it to a single dimension in the frequency axis before the LSTM layers handle the temporal aspects. Table 1 presents the architecture of the implemented CNN-LSTM model in this study, where the number of CNN layers is determined by the input dimension along the frequency axis, denoted as f in the table. The input shape is (1, 256, 256) for SPWVD and STFT, whereas it is (1, 48, 256) for SPWVD-MFCC and MFCC. Previous studies [16, 18, 37] have shown that CNN-LSTM models perform well in sound classification tasks.

MobileViT-s, developed by Mehta and Rastegari [38], is a hybrid model that combines CNN with Vision Transformer (ViT) architectures, allowing it to capture local details and

Table 1 Details of the CNN-LSTM

Layer	Hyperparameters	Output shape
Input (Feature segment)	–	$(1, f, 256)$
Conv-1	Output channel = $512/2^{N-1}$; Kernel size = (3, 1); Stride = (2, 1); Padding = (1, 0); Norm = Batch Norm; Activation = Leaky ReLU	$(32, f/2, 256)$
Conv-2	Output channel = $512/2^{N-2}$; Kernel size = (3, 1); Stride = (2, 1); Padding = (1, 0); Norm = Batch Norm; Activation = Leaky ReLU	$(64, f/2^2, 256)$
…	…	…
Conv-(N-1)	Output channel = 256; Kernel size = (3, 1); Stride = (2, 1); Padding = (1, 0); Norm = Batch Norm; Activation = Leaky ReLU	$(256, f/2^{N-1}, 256)$
Conv-N	Output channel = 512; Kernel size = $(f/2^{N-1}, 1)$; Stride = 1; Padding = 0; Norm = Batch Norm; Activation = Leaky ReLU	(512, 1, 256)
Reshape and Swap axes	–	(256, 512)
LSTM-1	Hidden size = 128	(256, 128)
Dropout	Dropout probability = 30%	(256, 128)
LSTM-2	Hidden size = 128	(256, 128)
Data select	–	(128)
BN1D	Feature number = 128	(128)
FC	Output feature = 10; Activation = Softmax	(10)
Output	–	(10)

global contextual information. ViT, introduced by Dosovitskiy et al. [39], processes images by dividing them into patches and treating these patches as sequences. MobileViT-s is designed with efficient modules and parameter optimisations to achieve high performance in tasks such as image classification, object detection, and semantic segmentation while maintaining low computational and memory costs.

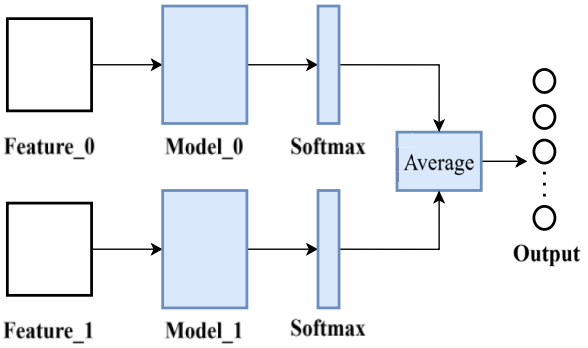


Fig. 6 Structure of the dual-stream classifier

Its effectiveness has been validated in various image-related studies [40–42].

Given the demonstrated benefits of multi-stream structures in enhancing classification accuracy [19, 20, 43], this study also constructs dual-stream models by identifying the top two features suitable for each classifier. The dual-stream approach, illustrated in Fig. 6, involves feeding two distinct features into separate streams of the model. The outputs are then combined in a Softmax layer to calculate class probabilities, which are averaged to produce the final prediction, leveraging the strengths of both streams rather than relying on a single prediction.

In conclusion, this paper employs ResNet50, CNN-LSTM, and MobileViT-s as benchmark classifiers to evaluate the performance of STFT, SPWVD, and their derived MFCC features for robust urban sound classification. By comparing these diverse architectures, the study aims to identify the most robust model for handling the complexities of urban sound data. Using dual-stream models further explores the potential of combining complementary feature sets to enhance classification accuracy. Overall, this approach provides a comprehensive evaluation of advanced architectures in urban sound classification, setting a foundation for further improvements in noise-robust sound monitoring systems.

3.4 Experiment Settings

For the experimental setup, ResNet50, CNN-LSTM, and MobileViT-s are used to classify features derived from STFT and SPWVD on the UrbanSound8K dataset, with further verification using additional datasets. All models were implemented without pre-training: ResNet50 was sourced from the "torchvision" module [44], MobileViT-s from "timm" [45], and CNN-LSTM was constructed using the "torch" module [46]. Modifications were made to adapt the models for urban sound classification by adjusting the input channels to 1 and output classes to 10, as ResNet50 and MobileViT-s were initially designed for 3-channel images and 1000 classes. Table 2 details the models' architecture, including

Table 2 Details of the models

	ResNet50	CNN-LSTM		MobileViT-s
Convolutional layers	49	5	8	35
Fully connected layers	1			
Other key layers	0	2 × LSTM		3 × Transformer
Number of parameters	23.50 M	0.99 M	0.86 M	4.94 M
FLOPs with an input shape of (1, 256, 256)	5.31 G	–	0.17 G	1.89 G
FLOPs with an input shape of (1, 48, 256)	1.05 G	0.24 G	–	0.38 G

the number of convolutional and fully connected layers, key components, model parameters, and FLOPs for processing time–frequency distributions (shape (1, 256, 256)) and their derived MFCC (shape (1, 48, 256)).

The training process was conducted over up to 100 epochs using the Adam optimiser with a batch size of 256 and an initial learning rate of 0.001. After each epoch, the model's performance was evaluated on a validation set, and the best-performing model was saved. If the training accuracy did not improve for ten consecutive epochs, training reverted to the best model, and the learning rate was reduced by a factor of ten to mitigate overfitting. Early stopping was employed if no validation accuracy improvement was observed for five successive learning rate reductions or if the learning rate fell below 1e-12. The final model was trained for one additional epoch using the training and validation sets to maximise data utilisation. All training and testing were performed on an NVIDIA A100 GPU with 80 GB of memory provided by the New Zealand eScience Infrastructure (NeSI).

4 Experiment Results and Discussion

This section introduces the experiments conducted in this study and the corresponding results. Experiments introduced in Sects. 4.1, 4.2, and 4.3 are based on UrbanSound8K, while Sect. 4.4 is based on the proposed UrbanSoundPlus. Additionally, Sect. 4.5 compares the proposed system with other studies on UrbanSound8K and DCASE2016. Finally, limitations are discussed in Sect. 4.6.

4.1 Compare the Proposed SPWVD-MFCC with SPWVD, STFT, and STFT-MFCC

In this experiment, ResNet50, CNN-LSTM, and MobileViT-s are employed to compare the classification accuracies of using SPWVD-MFCC with the benchmark SPWVD, STFT, and STFT-MFCC. The features are extracted with 512 frequency bins and a step length of 32 samples.

Table 3 collects the results of the comparison. Based on the results, we can draw the following conclusions: First, the average accuracy of STFT is consistently lower than the SPWVD regardless of the classifier used. In contrast, the improvements can be up to 24.4% while using MobileViT-s. Second, the average results of using SPWVD-MFCC consistently outperform the STFT-MFCC, with an improvement of up to 25.9% on ResNet50. The STFT-MFCC consistently achieves top accuracies for clean sound classification, exemplified by ResNet50 utilising STFT-MFCC, achieving a 95.4% accuracy rate. This high performance has led to its extensive use in existing studies. However, the significantly lower average accuracy of STFT-MFCC in noisy environments highlights that SPWVD-MFCC offers substantially better robustness under such conditions. Third, comparing the STFT and SPWVD with their derived MFCC reveals that the derived MFCC consistently achieves higher accuracy for clean sounds and average accuracy. This indicates that additional feature extraction steps, such as mapping the original linear frequency bins to their respective scaled frequencies and incorporating dynamic features (delta and delta-delta), are necessary to improve the signal's representativeness. Finally, we found that ResNet50 and MobileViT-s perform better than CNN-LSTM in processing spectrograms. In contrast, the three models have similar accuracies when using MFCC features. This suggests that the CNN mechanism in CNN-LSTM is less effective at processing the 256-dimensional frequency axis information in spectrograms than the 48-dimensional data in MFCC.

4.2 Compare SPWVD-MFCC with Different Step Lengths

The previous experiment demonstrated that SPWVD-MFCC significantly outperformed SPWVD, STFT, and STFT-MFCC in urban sound classification under noisy conditions. This section extends the experiment by adjusting the step length of SPWVD-MFCC with varying step lengths. The step lengths tested in this experiment are 16, 32, 64, 128, and 256 samples, while the number of frequency bins is fixed at 512.

Table 3 Classification accuracy of STFT, SPWVD, and their derived MFCC with a step of 32 samples

Classifier	Feature	SNR						Clean (%)	Avg (%)
		0 dB (%)	4 dB (%)	8 dB (%)	12 dB (%)	16 dB (%)	20 dB (%)		
ResNet50	STFT	23.2	33.6	44.4	50.5	54.8	57.0	58.5	46.0
	SPWVD	32.4	44.6	59.0	64.9	68.0	69.8	69.9	58.4
	STFT-MFCC	16.8	21.5	31.3	42.0	52.1	59.4	95.4	45.5
	SPWVD-MFCC	30.6	50.0	72.0	82.0	86.5	88.3	90.3	71.4
CNN-LSTM	STFT	16.2	18.5	22.2	26.3	28.0	29.0	30.5	24.4
	SPWVD	20.2	26.8	38.3	45.8	49.9	51.3	52.0	40.6
	STFT-MFCC	16.1	24.1	34.0	43.4	53.3	62.1	93.0	46.6
	SPWVD-MFCC	30.4	48.1	68.7	80.8	85.2	87.6	88.0	69.8
MobileViT-s	STFT	18.5	24.4	33.2	38.7	43.3	47.1	50.5	36.5
	SPWVD	29.5	46.5	63.0	69.1	71.8	72.9	73.3	60.9
	STFT-MFCC	18.7	25.5	36.3	44.9	54.3	63.3	92.3	47.9
	SPWVD-MFCC	30.0	50.8	69.3	79.7	83.2	84.6	85.4	69.0

Table 4 Classification accuracy of using SPWVD-MFCC with various step lengths

Classifier	Step length	SNR						Clean (%)	Avg (%)
		0 dB (%)	4 dB (%)	8 dB (%)	12 dB (%)	16 dB (%)	20 dB (%)		
ResNet50	16	27.6	44.1	65.4	78.1	84.3	87.0	88.3	67.8
	32	30.6	50.0	72.0	82.0	86.5	88.3	90.3	71.4
	64	29.0	47.5	68.2	81.4	86.5	87.9	87.5	69.7
	128	32.5	48.3	69.2	79.8	84.4	85.7	86.9	69.6
	256	27.7	43.5	60.2	72.4	76.9	79.3	80.6	62.9
CNN-LSTM	16	29.7	47.2	66.7	76.7	82.8	85.4	87.5	68.0
	32	30.4	48.1	68.7	80.8	85.2	87.6	88.0	69.8
	64	29.3	46.8	67.9	79.8	85.4	87.4	88.3	69.3
	128	26.8	47.0	66.3	78.9	84.2	85.9	87.0	68.0
	256	29.3	46.6	65.4	76.0	80.4	82.2	84.4	66.3
MobileViT-s	16	31.5	48.3	66.7	76.7	80.9	82.8	84.3	67.3
	32	30.0	50.8	69.3	79.7	83.2	84.6	85.4	69.0
	64	29.2	46.5	64.6	75.2	79.1	81.4	82.7	65.5
	128	33.1	50.2	66.0	73.5	78.0	79.6	80.4	65.8
	256	30.2	45.4	60.8	69.7	73.7	75.1	76.9	61.7

Table 4 presents the detailed results of this experiment. ResNet50 achieved the highest accuracy using SPWVD-MFCC with a step length of 32 samples, reaching 71.4% on average and 90.3% on clean sound. The highest average accuracy of CNN-LSTM using SPWVD-MFCC is 69.8%, achieved with 32 samples. MobileViT-s also obtains the highest accuracy at 69% using the 32-sample step-length SPWVD-MFCC, showing that the step length of 32 samples is the most proper for the classifiers. Furthermore, as the step length increased to 128 or 256 samples, the accuracies of ResNet50 and MobileViT-s decreased faster compared to CNN-LSTM. This indicates that the reduction in temporal resolution impacts ResNet50 and MobileViT-s more substantially than CNN-LSTM. Given that the CNN-LSTM specifically processes temporal information using LSTM, it can efficiently handle the relationships across timestamps, leading to a good understanding of the data even when the temporal resolution is low. In conclusion, the best average accuracies of ResNet50, CNN-LSTM, and MobileViT-s are obtained using 32-sample step-length SPWVD-MFCC, reaching 71.4%, 69.8%, and 69.0%, respectively.

Table 5 Classification accuracy of dual-stream structures based using SPWVD-MFCC with a step length of 32 samples

Dual-stream classifier	SNR						Clean (%)	Avg (%)
	0 dB (%)	4 dB (%)	8 dB (%)	12 dB (%)	16 dB (%)	20 dB		
ResNet50-CNN-LSTM	31.7	51.5	74.0	86.2	90.0	91.8	92.1	73.9
ResNet50-MobileViT-s	30.4	51.1	74.3	83.9	87.5	88.9	90.3	72.3
CNN-LSTM-MobileViT-s	31.1	51.1	72.7	84.4	87.5	88.7	89.3	72.1

4.3 Performance of the Dual-Stream Structure

Table 5 compares the accuracy of dual-stream classifiers based on ResNet50, CNN-LSTM, and MobileViT-s. Feature selection for this section is informed by previous experimental results, where a 32-sample step-length SPWVD-MFCC provided the best results across all three classifiers. The dual-stream ResNet50-CNN-LSTM achieves the highest average accuracy and the best performance in classifying clean audio, with 73.9% and 92.1%, respectively. Compared to the single-stream ResNet50 and CNN-LSTM models in Table 4, the dual-stream structure improves classifying clean audio and WGN-mixed audio across all SNRs, demonstrating its ability to leverage both streams' strengths effectively.

Figure 7 illustrates the averaged confusion matrices of ResNet50, CNN-LSTM, and the dual-stream ResNet50-CNN-LSTM across various SNRs. Labels 0 to 9 correspond to Air conditioners, Car horns, Children playing, Dog bark, Drilling, Engine idling, Gunshots, Jackhammer, Sirens, and Street music. The matrices indicate that the dual-stream structure provides improved or similar accuracies in seven out of ten classes. For the remaining three classes, namely Dog Bark, Drilling, and Sirens, the dual-stream structure offers accuracies between the two single streams but is closer to the higher one. Thus, the dual-stream structure improves classification accuracy over the single-stream models by leveraging their respective strengths, and the dual-stream ResNet50-CNN-LSTM has the best accuracy compared to other models.

4.4 Test the Proposed System on UrbanSoundPlus

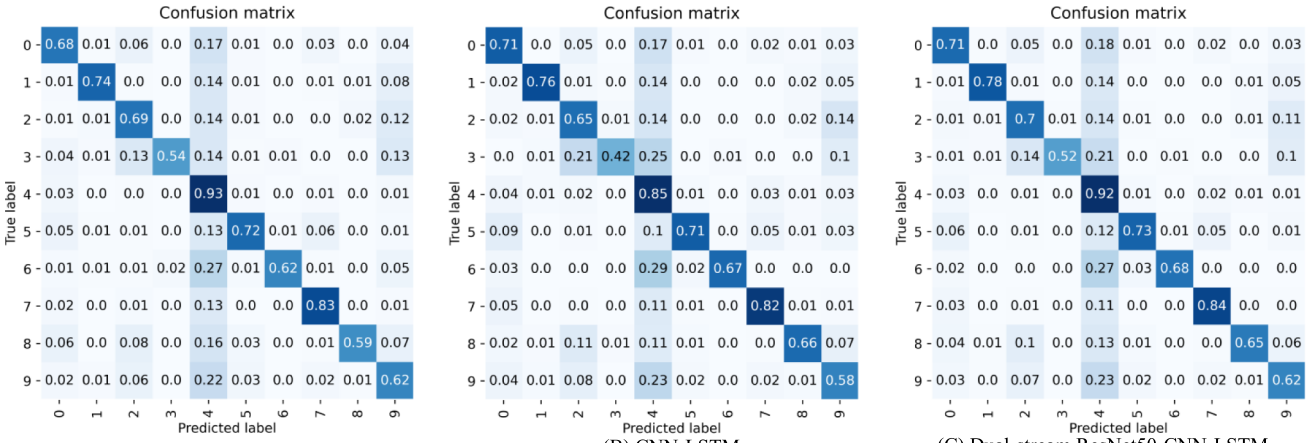
This experiment compares the classification systems on the newly introduced dataset, UrbanSoundPlus, to show the generalisation of the proposed methods. ResNet50, CNN-LSTM, and MobileViT-s are employed to classify STFT- and SPWVD-MFCC with a step length of 32 samples to compare with the dual-stream ResNet50-CNN-LSTM using 32-sample step-length SPWVD-MFCC. As shown in Table 6, the results align with previous experiments. While classifiers using STFT-MFCC achieve higher accuracy in clean sound classification, all classifiers using the proposed SPWVD-MFCC demonstrate superior average accuracy through the SNRs, surpassing the STFT-MFCC for up to 37.2% on average accuracy (ResNet50 using SPWVD-MFCC versus using STFT-MFCC).

The dual-stream ResNet50-CNN-LSTM achieves the top average accuracy of 70.1%, surpassing the single-stream ResNet50, CNN-LSTM, and MobileViT-s by 1.4%, 6.8%, and 6.8%, respectively. Comparing the results with those on UrbanSound8K, a decline in all corresponding accuracies is observed. This means the proposed UrbanSoundPlus, which includes a broader range of urban sounds, is more complex and challenging than UrbanSound8K. This complexity benefits the system's classification accuracy in real-world environments. In conclusion, the dual-stream ResNet50-CNN-LSTM using 32-sample step-length SPWVD-MFCC has better average accuracy than other approaches on the proposed UrbanSoundPlus.

4.5 Comparison to Other Studies

This study employs dual-stream ResNet50-CNN-LSTM using the proposed SPWVD-MFCC to classify urban sounds in noisy environments. Table 7 lists the clean sound classification accuracy obtained by the proposed and existing systems on the UrbanSound8K dataset without data augmentation, where the proposed system reached 92.1% accuracy. The performance in the middle tier of the comparison list was anticipated, as our experimental results indicate that SPWVD-MFCC does not excel in classifying clean sounds compared to STFT-MFCC. However, it is crucial to highlight that testing urban sound classification systems only on clean audio does not accurately reflect their performance in real-world deployments, as the actual environments are usually noisy.

To the best of our knowledge, no prior study has reported the accuracy of classifying WGN-mixed UrbanSound8K data. We conducted two additional experiments to benchmark our system against others and compare the classification accuracy in noisy environments. In the first additional experiment, we replicated the methodology proposed by Hinrichs et al. in study [23]. Their approach has tried to extract MFCC features from STFT, Stockwell, WVD, and STFT-filtered WVD under different configurations. These eight features (as detailed in Table 8) were separately fed into eight CNNs



(A) ResNet50 (B) CNN-LSTM (C) Dual-stream ResNet50-CNN-LSTM

Fig. 7 Averaged confusion matrices of the dual-stream ResNet50-CNN-LSTM using 32-sample step-length SPWVD-MFCC

Table 6 Classification accuracy on UrbanSoundPlus using 32-sample step-length STFT- and SPWVD-MFCC

Dual stream	Classifier	Feature	SNR						Clean (%)	Avg (%)
			0 dB (%)	4 dB (%)	8 dB (%)	12 dB (%)	16 dB (%)	20 dB (%)		
No	ResNet50	STFT-MFCC	8.3	9.2	14.8	22.1	30.8	42.4	93.3	31.5
		SPWVD-MFCC	25.7	48.9	68.5	78.5	84.3	86.6	88.2	68.7
	CNN-LSTM	STFT-MFCC	8.2	9.7	13.6	20.9	30.0	42.9	91.3	30.9
		SPWVD-MFCC	23.8	44.1	62.0	71.4	77.3	80.8	83.9	63.3
	MobileViT-s	STFT-MFCC	8.5	10.9	15.9	25.3	37.1	45.4	91.0	33.4
		SPWVD-MFCC	25.8	44.4	61.0	71.8	78.1	80.4	81.8	63.3
Yes	ResNet50 CNN-LSTM	SPWVD-MFCC	26.2	51.5	70.1	79.6	85.5	87.9	89.8	70.1

Table 7 Comparison of the accuracy obtained by the proposed and existing systems using deep-learning models on clean audio from UrbanSound8K without data augmentation

Study	Network	Feature	Accuracy (%) on clean audio
[47]	SqueezeNet18	STFT	80.1
[48]	CNN-BiLSTM-Attention	STFT-MFCC	80.2
[43]	D-2-DensNet	STFT-MFCC, STFT-GFCC	84.83
[49]	ResNet-50	Log-Mel-STFT	86.49
[50]	CNN	STFT	88
[51]	PSO-optimised CNN	Scalogram	91.17
Proposed	Dual-stream ResNet50-CNN-LSTM	SPWVD-MFCC	92.1
[52]	LSTM	STFT-MFCC, Chroma-STFT	93.76
[16]	CNN-LSTM (ours)	STFT-MFCC	94.94
[20]	CNN	log-Mel-STFT, Raw audio	94.97
[19]	TSCNN-DS	Chroma, Spectral Contrast, Tonnetz, Log-Mel-STFT, STFT-MFCC	97.2

Table 8 Classification accuracies of the proposed and the replicated existing system on WGN-mixed UrbanSound8K

Study	Network	Concatenated features	SNR						Clean (%)	Avg (%)
			0 dB (%)	4 dB (%)	8 dB (%)	12 dB (%)	16 dB (%)	20 dB (%)		
[23] (replicated)	Multi-stream CNN (Post-mix)	STFT ($n = 512$)-MFCC STFT ($n = 1024$) -MFCC STFT ($n = 2048$)-MFCC Stockwell-MFCC WVD-MFCC Wigner–Ville Filtered ($n = 256$)-MFCC Wigner–Ville Filtered ($n = 512$)-MFCC Wigner–Ville Filtered ($n = 1024$)-MFCC	28.5	43.9	63.0	76.3	84.4	88.9	94.4	68.5
Proposed	Dual-stream ResNet50-CNN-LSTM	SPWVD-MFCC (One feature)	31.7	51.5	74.0	86.2	90.0	91.8	92.1	73.9

with similar structures, and the resulting outputs were then concatenated and processed through a feed-forward network. We replicated their approach on the same dataset used in this study to ensure a fair and convincing comparison, because they only conducted tests based on a speech command recognition dataset. The results, shown in Table 8, indicate that our proposed method using SPWVD-MFCC consistently outperformed the replicated system across all SNR levels in noisy environments. Notably, the most significant performance gap occurred between 4 and 12 dB, where our system demonstrated approximately a 10% improvement. Although the replicated system exhibited a 2.3% advantage in classifying clean sound, our system achieved an average accuracy of 5.4% higher, underscoring its superior noise resistance and robustness for real-world applications. This is primarily attributed to the superior time–frequency feature extraction and information fusion strategy of the proposed classification system. Compared to the method introduced by Hinrichs et al., our approach effectively eliminates information redundancy and feature conflicts, enhancing stability under low-SNR conditions.

In the second additional experiment, we employed another environmental sound dataset, DCASE2016, to further compare the proposed system with that introduced by Huang et al., who used WGN-mixed DCASE2016 data to simulate noisy environments [43]. DCASE2016 [53] is a dataset designed for the Detection and Classification of Acoustic Scenes and Events ((DCASE) challenge, containing 15 categories. To align with the testing conditions of the comparison system, we utilised the original 30-s audio segments from DCASE2016, sampled at 44.1 kHz, with adjustments such as setting the step length to 8192 and the number of MFCC coefficients to 56. Consequently, the dimensions of the SPWVD-MFCC for this experiment were (1, 174, 162), meaning each 30-s audio segment was converted into 162 timestamps, with each timestamp comprising 174 MFCC coefficients that include both dynamic and static features. The comparison results are presented in Table 9. Our system demonstrated a significant advantage in accuracy on WGN-mixed audio, outperforming the compared system by 7.6% on average, achieving an overall accuracy of 82.0%. Notably, the proposed system maintained similar accuracies across SNR levels of 20 dB, 30 dB, 40 dB, and 50 dB, and under clean conditions, with a narrow accuracy variation of just 2.7%, significantly lower than the 15.5% variation observed in the compared system. This consistency suggests that the performance of the proposed system is largely unaffected by noise when the SNR exceeds 20 dB, indicating that it can effectively learn from the clean signal and handle scenarios with an SNR of 20 dB or higher significantly better than the approach proposed by Huang et al. At 10 dB, our system achieved a 3% advantage over the compared system. However, this advantage was less pronounced than higher SNR levels, indicating that classifying WGN-mixed DCASE2016 audio at SNRs below 20 dB is notably more challenging. This could be attributed to pre-existing noise in some DCASE2016 audio segments, where adding WGN further complicates effective feature extraction, leading to a marked decrease in accuracy at 10 dB SNR.

4.6 Limitations

While our study highlights the robustness of the dual-stream classifiers with the proposed SPWVD-MFCC feature, several limitations warrant attention in future research. Firstly, some fixed settings were used for feature extraction due to time constraints. Future work should explore a broader range of parameters to optimise WVD feature settings specifically for urban sound classification. Secondly, the current study only mixed WGN with clean urban sounds, which provides uniform energy across frequencies. To better assess the performance of WVD-derived features, future studies should incorporate a variety of noise types to evaluate their robustness under diverse noise conditions. Thirdly, this study did not explicitly balance the class distribution in the newly introduced UrbanSoundPlus dataset, leading to potential biases during training and testing. Future research should address this issue by employing appropriate techniques, such as data augmentation, re-sampling, or class-weighted loss functions, to mitigate the impact of class imbalance on model performance. Lastly, the computational cost of extracting SPWVD features was significantly higher than that of STFT, likely due to the autocorrelation and smoothing processes involved. Addressing this time efficiency gap is essential for enabling real-time classification with SPWVD-MFCC in noisy environments, making it more practical for real-world applications.

5 Conclusion

This paper focuses on improving the robustness of urban sound classification in noisy environments. Using UrbanSound8K and the newly introduced UrbanSoundPlus datasets, we compared the accuracies of various feature-classifier combinations. Comprehensive experiments on clean audio and WGN-mixed audio with SNRs ranging from 0 to 20 dB showed that the proposed SPWVD-MFCC demonstrated superior average accuracies, improving up to 37.2% over traditional STFT-MFCC. This showcases the effectiveness of the proposed feature in enhancing classification robustness. The proposed dual-stream ResNet50-CNN-LSTM using SPWVD-MFCC with a step length of 32 samples achieved the highest average accuracies of 73.9% on UrbanSound8K and 70.1% on UrbanSoundPlus. Evaluations on DCASE2016 yielded an average accuracy of

Table 9 Classification accuracies of the proposed and the existing system on WGN-mixed DCASE2016

Study	Network	Concatenated features	SNR					Clean (%)	Avg (%)
			10 dB (%)	20 dB (%)	30 dB (%)	40 dB (%)	50 dB (%)		
[43]	D-2-DensNet	STFT-MFCC STFT-GFCC	68.3	69.7	72.4	74.8	77.2	85.2	74.4
Proposed	Dual-stream ResNet50-CNN-LSTM	SPWVD-MFCC	71.3	82.4	84.2	84.2	84.9	84.9	82.0

82.0% on clean audio and WGN-mixed audio with SNRs ranging from 10 to 50 dB, outperforming the study testing with WGN-mixed audio by 7.6%. These outstanding achievements across the three datasets affirm the generalisation capability of the proposed feature and model in robust urban sound classification under noisy conditions.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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